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Research Project Report on the Topic:
Guitar Music Transcription Based on an Audiovisual Signal

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Annotation

Automatic Guitar Transcription (AGT) faces significant challenges due to string-fret ambiguity, where identical pitches can be played at multiple positions. Existing audio-only methods often fail to recover accurate fingering, limiting their practical utility for musicians. This work proposes a novel multimodal deep learning architecture that integrates audio spectrograms with visual data from performance videos to resolve these ambiguities. By leveraging visual cues regarding finger placement and string activation, the model directly estimates string-fret assignments rather than relying solely on pitch estimation and heuristic rules. The architecture employs MIDI pitch regression rather than fret classification, enabling adaptation to alternative tunings without model retraining. Experimental results show a 34% relative improvement in onset F1-score ($0.309 \rightarrow 0.414$) compared to audio-only baselines on a custom dataset. This provides a foundation for automated music education tools and performance assessment systems.

Аннотация

Автоматическая транскрипция гитарной музыки сталкивается с серьёзными трудностями из-за позиционной неоднозначности: одинаковые звуки могут быть извлечены на разных струнах и ладах. Существующие методы, использующие только аудиосигнал, часто ошибаются в определении аппликатуры, что ограничивает их практическую применимость для музыкантов. В данной работе предлагается новая мультимодальная архитектура глубокой нейронной сети, объединяющая аудиоспектрограммы и видеоданные исполнительских записей для устранения этой неоднозначности. Используя визуальные признаки положения пальцев и активации струн, модель напрямую оценивает пары «струна-лад», а не только основывается на высоте тона и эвристических правилах. Архитектура применяет регрессию MIDI-нот вместо классификации ладов, что обеспечивает адаптацию к альтернативным строям без переобучения модели. Экспериментальные результаты показывают относительное улучшение F1-меры для детектирования атак на 34% ($0.309 \rightarrow 0.414$) по сравнению с аудиобазовой моделью на собственном наборе данных. Это создаёт основу для автоматизированных систем обучения музыке и оценки исполнения.

Keywords

Automatic Music Transcription, Guitar Tablature, Multimodal Learning, String-Fret Ambiguity, Alternative Tunings, Onset Detection, Pitch Regression, Audio-Visual Fusion.

1 Introduction

The guitar is a highly expressive yet mechanically complex instrument. Guitarists employ a wide range of techniques—including hammer-ons, pull-offs, tapping, and natural or artificial harmonics—that involve intricate interactions between finger placement, string excitation, and timing [15]. These nuances pose significant challenges for automatic transcription systems.

Historically, guitar music has been notated using two primary formats: standard sheet music and tablature (tabs). Sheet music provides a universal, standardized representation that encodes both pitch and rhythm. However, it does not specify how a note should be played on the guitar (i.e., which string and fret to use).

In contrast, tablature explicitly indicates string and fret positions, making it far more intuitive and practical for guitarists [21]. As a result, most guitar-focused AMT systems aim to generate tablature rather than conventional notation [1, 17, 29, 19].

As illustrated in Figure 1.1, guitar tabs are commonly represented in a simple ASCII format. Notably, such representations often omit explicit rhythmic information. While this is generally acceptable to experienced players—who can infer timing from context—the core objective of guitar transcription remains the accurate identification of which string and fret produce each note.

The figure displays three musical representations of J.S. Bach's Bourrée, organized into three rows. The first row shows standard musical notation on a five-line staff, featuring various note values, accidentals, and articulation marks like 'i' (accents) and 'm' (marcato). The second row shows traditional guitar tablature, with numbers 0-4 placed on a six-line staff to indicate fret positions. The third row shows ASCII tablature, using a similar six-line staff but with numbers and dashes to represent fret positions in a text-based format. A 'B II' marking with a repeat sign is present above the second measure of the first two rows.

Figure 1.1: Standard notation, Tablature, ASCII Tablature comparison. J.S. Bach Bourrée [5].

That being said, more informative tablature would ideally include temporal annotations. In practice, however, precise note duration modeling is less critical for the guitar than for instruments

like the piano. The guitar typically has only six strings, and notes are often terminated by the onset of subsequent notes due to limited sustain and the physical necessity of repositioning fingers quickly. Moreover, guitarists frequently allow notes to ring as long as musically appropriate, meaning that actual note durations are highly dependent on individual playing style and artistic interpretation—not just the underlying score. Notably, this offset ambiguity is recognized across the AMT community: Benetos et al. [6] explicitly state that “note offsets can be ambiguous” even for piano transcription, leading many evaluation frameworks to prioritize onset and pitch accuracy over precise offset timing. Thus, for guitar transcription specifically, systems that omit exact offset times can still yield musically useful results.

Rhythmic expression further complicates the task. As noted in [22], certain stylistic elements—such as swing feel—are rarely encoded in standard notation but are essential to authentic performance. Similarly, dynamics, articulation, and timbral variations are often conveyed through marginal annotations or performer intuition rather than explicit symbols. While the cited work addresses the inverse problem (generating expressive performances from scores), it highlights a shared challenge between music generation and AMT: the gap between symbolic notation and acoustic reality.

For guitar AMT, the highest priority is therefore the reliable and efficient recovery of essential musical content: namely, pitch, string, and fret. Secondary attributes such as dynamics or precise note offsets may be addressed in future work.

A fundamental challenge unique to stringed instruments is positional ambiguity: a single pitch can often be played on multiple strings and at different frets (see Figure 1.2) [24]. Consequently, even a correctly identified pitch may correspond to several valid tablature representations. This ambiguity is exacerbated by advanced playing techniques and, critically, by non-standard tunings. Many classical and contemporary pieces employ alternate tunings (e.g., Drop D, Open G, DADGAD), and modern guitarists frequently experiment with custom configurations [28]. An ideal transcription system should be robust to such variations without requiring retraining or manual intervention.

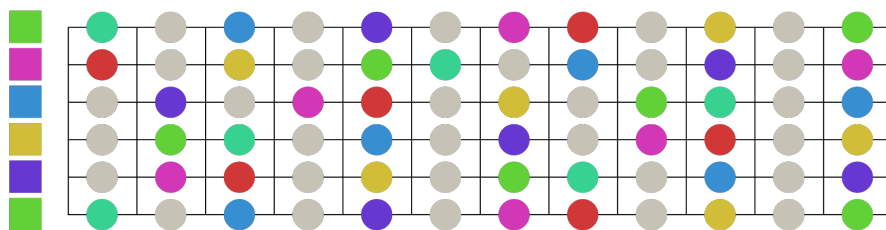


Figure 1.2: Guitar fretboard with notes.

Current state-of-the-art guitar transcription systems predominantly rely on frame-level piano-roll representations and audio-only inputs. While computationally convenient, this formulation requires precise frame-wise annotations of note onsets and offsets—yet offset timing in guitar performance is often ambiguous and stylistically variable. Furthermore, audio spectrograms provide reliable pitch information but lack direct evidence about which string produced a given note, as different strings playing the same pitch produce nearly identical frequency content. This fundamental limitation forces current systems to rely on statistical priors for string assignment, which frequently fail in polyphonic passages or non-standard tunings.

In contrast, visual signals contain explicit spatial cues: the fretting hand reveals precise finger positions, while the plucking hand indicates string activation order. Despite this complementary information, few AMT systems have leveraged video data. This work proposes a novel multimodal architecture that fuses audio spectrograms with close-up video footage to directly observe string selection. By incorporating visual evidence, our approach reduces the string-fret ambiguity inherent in audio-only methods and enables robust transcription across alternative tunings without requiring model retraining.

2 Related Work

Given the unique challenges of guitar tablature transcription (GTT), several approaches have been proposed. Early methods relied on traditional audio signal processing [16] or probabilistic modeling [32]. More recently, deep neural networks [23] have emerged as the dominant paradigm due to their ability to learn complex mappings from audio to symbolic representations.

Among these, TabCNN [29] is widely regarded as the state-of-the-art for GTT. Unlike traditional pitch transcription systems [6, 13], TabCNN is explicitly designed to predict fingering positions—i.e., string-fret pairs—enabling direct generation of guitar tablature. The model processes Constant-Q Transform (CQT) spectrograms [7], which provide a musically meaningful frequency representation by aligning bins with musical pitches. Each time frame is treated as an image-like input to a convolutional neural network (CNN), producing a 6×21 output matrix representing softmax probabilities over frets for each of the six strings. The authors also introduce evaluation metrics tailored to GTT, acknowledging that while their system operates at the frame level, its outputs can be post-processed into note-level transcriptions.

Subsequent work has shifted toward note-level and tokenized representations. Kim et al. [19] extended the TabCNN architecture by incorporating self-attention [27] mechanisms and beat-aligned temporal context, enabling direct estimation of discrete note events. This approach

demonstrated competitive performance not only in note-level transcription but also in frame-level metrics. In a follow-up study [20], the same authors adopted a tokenized output format, inspired by modern automatic speech recognition (ASR). Their sequence-to-sequence Transformer model employs Connectionist Temporal Classification (CTC) loss [11] to handle unaligned input–output sequences—a technique particularly valuable given the scarcity of precisely annotated guitar datasets. However, because the target representation is based on MIDI, which encodes pitch and timing but not string or fret information, the original fingering is irreversibly lost.

To address this limitation, recent work has explored fingering recovery from MIDI. The Fretting Transformer [12] uses an attention-based model to map MIDI notes to plausible string–fret assignments. While it produces musically valid tablature, it cannot guarantee fidelity to the performer’s original fingering, especially in passages with multiple valid fingerings.

As noted in Section 1, most existing guitar transcription systems—including TabCNN and even early token-based adaptations—ultimately rely on an intermediate or final frame-level piano-roll representation. While this formulation aligns naturally with spectrogram inputs and enables the use of standard losses like binary cross-entropy, it suffers from fundamental drawbacks. First, it requires precise frame-wise annotations of note onsets and, more problematically, offsets. This introduces significant label noise and increases annotation cost. Second, the piano-roll format conflates acoustic presence with symbolic intent: a ringing note may be acoustically present long after its notated duration, leading to mismatches between ground truth and performer interpretation.

Nevertheless, the most common limitation across all aforementioned methods is their reliance on audio-only inputs. Although audio provides reliable pitch information, it offers little direct evidence about which string produced a given note—a fundamental ambiguity in guitar transcription. Despite this, few AMT systems have leveraged video. Notably, Asmar [2] investigated vision-only GTT using RGB-D video and classical computer vision techniques—including object segmentation, optical flow, Hough transforms, and clustering. While the system underperformed compared to audio-based methods, it demonstrated that visual data alone can support meaningful transcription, particularly when audio is unavailable or degraded.

More recently, multimodal approaches combining audio and video have gained traction. TapToTab [10] proposes a pipeline that uses YOLO [26] to detect and segment the fretboard in video frames, then infers active fret regions via visual analysis, which are fused with audio-derived pitch estimates. The authors report improved fret prediction accuracy, underscoring the complementary nature of the two modalities.

A persistent bottleneck across all deep learning-based GTT systems is the lack of large-scale, high-quality datasets. Public audio datasets such as GuitarSet [31], AGPT [9], and IDMT-

SMT-Guitar [18] exist, but many researchers consider them insufficient for training modern architectures without extensive data augmentation or synthetic pretraining [33]. Synthetic audio—often generated from MIDI using simple oscillators—is commonly used to scale training, though it fails to capture realistic timbral and expressive nuances.

Critically, to the best of our knowledge, no public video datasets for GTT are currently available. This gap stems from the labor-intensive nature of annotation: accurate tablature requires precise labeling of both onsets and fret-string assignments, often necessitating specialized hardware (e.g., hexaphonic pickups) or expert manual review.

These challenges motivate our work: we seek to develop a data-efficient, multimodal GTT framework that (1) explicitly recovers string-fret assignments using visual cues, (2) minimizes reliance on precisely aligned onset and offset annotations, and (3) generalizes across alternative tunings—thereby addressing core limitations of existing approaches.

3 Proposed Architecture

This section details the architectural design of our multimodal guitar tablature transcription system. The proposed framework consists of three core components: (1) an audio branch for spectrogram processing, (2) a video branch for visual feature extraction, and (3) a fusion module that integrates both modalities via cross-attention. We first describe the audio-only baseline, then extend it to the full multimodal architecture.

3.1 Audio Baseline

Problem Formulation. Following the frame-level paradigm established by TabCNN, our model processes fixed-length temporal windows and produces per-string predictions for the center frame. However, unlike TabCNN—which performs multi-class classification over discrete fret positions (0–24)—we decouple pitch estimation from physical fingering. Specifically, for each of the six strings, the model predicts:

- 1 A binary onset activation $o_s \in [0, 1]$, indicating the probability that string s was plucked at the target frame;
- 2 A continuous MIDI pitch value $p_s \in [0, 1]$, representing the sounding note as a normalized regression target.

This design choice provides two key advantages. First, explicit onset modeling enables accurate temporal alignment without requiring precise offset annotations, which are notoriously

ambiguous in guitar performance [6]. Second, by regressing continuous MIDI pitch values rather than classifying discrete fret numbers, the model becomes inherently tuning-agnostic. Standard classification approaches output a fixed matrix of fret probabilities (e.g., 6×21), which implicitly assumes a specific physical layout of the fretboard—namely, standard tuning. If the tuning changes, the physical location of pitches shifts, rendering the learned class mappings invalid. In contrast, our pitch regression operates in a universal acoustic space; given a known tuning configuration as side information, the physical fret number is simply derived post-hoc via:

$$\text{fret}_s = \text{MIDI}_{\text{note},s} - \text{MIDI}_{\text{open_string},s} \quad (1)$$

Input Representation. The audio input is a Constant-Q Transform (CQT) spectrogram [7] computed with a sample rate of 22,050 Hz and a hop size of 512 samples (23.2 ms per frame). Each input consists of 13 consecutive frames (302 ms temporal context), providing sufficient local information for onset detection while maintaining computational efficiency. The frequency axis comprises 72 bins, encompassing the full guitar range and relevant harmonics. The input tensor has shape $[B, 1, 13, 72]$, where B denotes batch size.

Architecture. The audio branch employs a convolutional neural network with three processing blocks (Figure 3.1):

- **Block 1:** Two 3×3 convolutional layers with 64 filters, batch normalization, ReLU activation, followed by 2×2 max pooling. Output: $[B, 64, 6, 36]$.
- **Block 2:** Two convolutional layers (one with a dilation rate of 2) with an integrated Squeeze-and-Excitation (SE) block [14] for channel-wise feature recalibration, followed by max pooling. Output: $[B, 64, 3, 18]$.
- **Block 3:** Two convolutional layers with a residual connection [25], maintaining spatial dimensions $[B, 64, 3, 18]$.

Global average pooling reduces the feature map to $[B, 64]$, which is then projected to a 512-dimensional embedding via a fully connected layer with ReLU activation and dropout ($p = 0.3$). Following Hu et al. [14], a Squeeze-and-Excitation (SE) block is integrated into Block 2 to perform channel-wise feature recalibration. This mechanism adaptively weights channel importance, allowing the network to focus on more informative features. It adds negligible computational overhead (approximately 196K parameters total for the audio baseline) while aligning with best practices for lightweight CNN architectures.

Loss Function. The total loss combines binary cross-entropy (BCE) for onset detection and mean squared error (MSE) for pitch regression:

$$\mathcal{L} = \lambda_{\text{onset}} \mathcal{L}_{\text{BCE}}(\mathbf{o}, \hat{\mathbf{o}}) + \lambda_{\text{pitch}} \sum_{s=1}^6 o_s \cdot \mathcal{L}_{\text{MSE}}(p_s, \hat{p}_s) \quad (2)$$

where $\mathbf{o}$ and $\hat{\mathbf{o}}$ denote ground-truth and predicted onset probabilities, respectively. Crucially, the pitch loss is weighted by the onset indicator o_s , ensuring that the model is not penalized for pitch predictions on silent strings. We set $\lambda_{\text{onset}} = 1.0$ and $\lambda_{\text{pitch}} = 1.0$ by default, with an additional positive class weight of 2.0 for onset BCE to address class imbalance (onsets are sparse events).

Output. The final output consists of two vectors of dimension 6:

- Onset probabilities: $\hat{\mathbf{o}} \in [0, 1]^6$ (sigmoid activation)
- Normalized MIDI pitch: $\hat{\mathbf{p}} \in [0, 1]^6$ (linear activation)

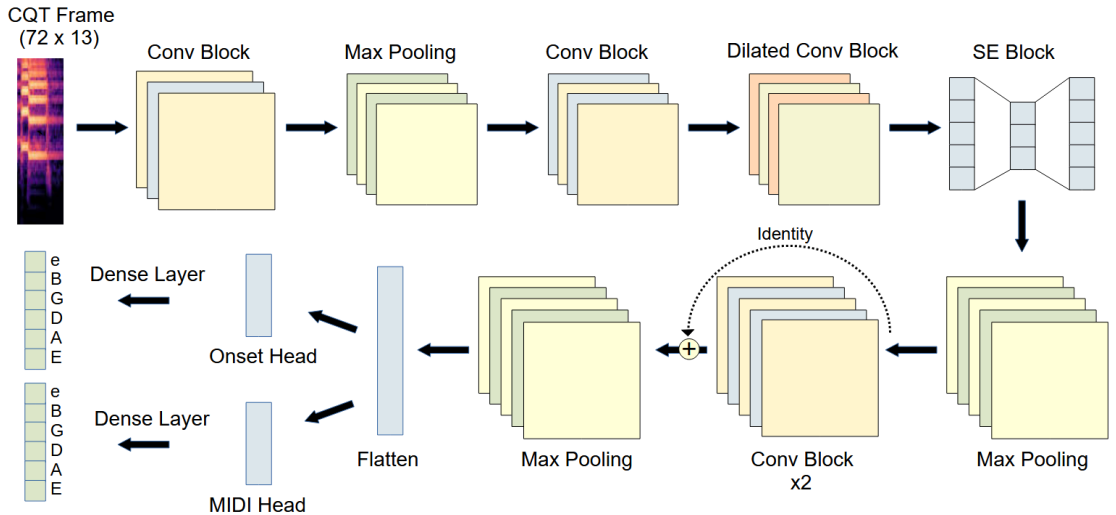


Figure 3.1: Audio branch architecture.

3.2 Multimodal Extension

Motivation for Visual Modality. As previously established, audio-only approaches face a fundamental limitation: different strings playing the same pitch produce nearly identical frequency content, creating inherent string-fret ambiguity. Prior work has explored video of the fretting hand (left hand) to identify finger positions [10]. However, this requires clear visibility of the fretboard and careful consideration of possible guitar tunings. In contrast, we propose using close-up video of the strings near the soundhole (picking hand region), where all six strings remain mostly visible throughout the performance. String vibration patterns and plucking hand

movements provide complementary spatial cues that guide the model in disambiguating which string produced each note.

Video Preprocessing. The dataset collection description is provided in Section 4. Original performance videos are recorded at 1080p resolution and then downsampled to reduce computational load while preserving string visibility. A fixed crop of 224×224 pixels is extracted from the soundhole region, ensuring that all six strings are visible within the frame. This crop size is chosen to match the input requirements of pretrained convolutional backbones while maintaining sufficient spatial resolution for string detection.

Video Branch Architecture. The video branch processes temporal sequences of 7 consecutive frames (280 ms at 25 fps), matching the temporal scale of the audio input (13 frames, 302 ms). The architecture comprises three stages (Figure 3.2):

- 1 **Frame Encoding:** Each frame is processed independently by a ConvNeXt V2 Atto backbone [30], a lightweight convolutional architecture pretrained on ImageNet [8]. The backbone produces 320-dimensional feature vectors per frame, yielding a sequence of shape $[B, 7, 320]$. The backbone is initially frozen during training to prevent overfitting on small datasets, with optional unfreezing in later training stages.
- 2 **Temporal Encoding:** A Temporal Convolutional Network (TCN) [26] processes the sequence of frame features to capture motion patterns indicative of plucking events. The TCN consists of three 1D convolutional layers with a kernel size of 3 and dilation rates $\{1, 2, 4\}$, providing a receptive field of exactly 7 frames. Batch normalization, ReLU activation, and dropout ($p = 0.2$) are applied after each convolution. Global temporal pooling aggregates the sequence into a single 320-dimensional vector.
- 3 **Projection:** A fully connected layer projects the 320-dimensional video embedding to 512 dimensions, matching the audio branch output space.

Modality Fusion via Cross-Attention. To integrate audio and visual information, we employ a cross-attention mechanism where audio features query video features (Figure 3.2). This design allows the model to selectively attend to visual cues that resolve ambiguities in the audio signal. The fusion module operates as follows:

- 1 **Video Projection:** The 512-dimensional video embedding is projected to match the audio embedding space via a linear layer.

- 2 **Multi-Head Cross-Attention:** Audio features serve as queries, while video features serve as keys and values in a multi-head attention layer [27] with 8 attention heads and dropout ($p = 0.1$). This produces an attended video representation aligned with the audio features.
- 3 **Residual Connection + Layer Norm:** A residual connection [25] and layer normalization [4] stabilize training and preserve audio information.
- 4 **Output Projection:** The original audio embedding and attended video representation are concatenated and projected back to 512 dimensions via an MLP with ReLU activation and dropout.

Prediction Heads. The fused 512-dimensional representation is passed to two parallel prediction heads, identical to the audio-only baseline:

- **Onset Head:** MLP ($512 \rightarrow 256 \rightarrow 6$) with sigmoid activation, producing per-string onset probabilities.
- **Pitch Head:** MLP ($512 \rightarrow 256 \rightarrow 6$) with linear activation, producing normalized MIDI pitch values.

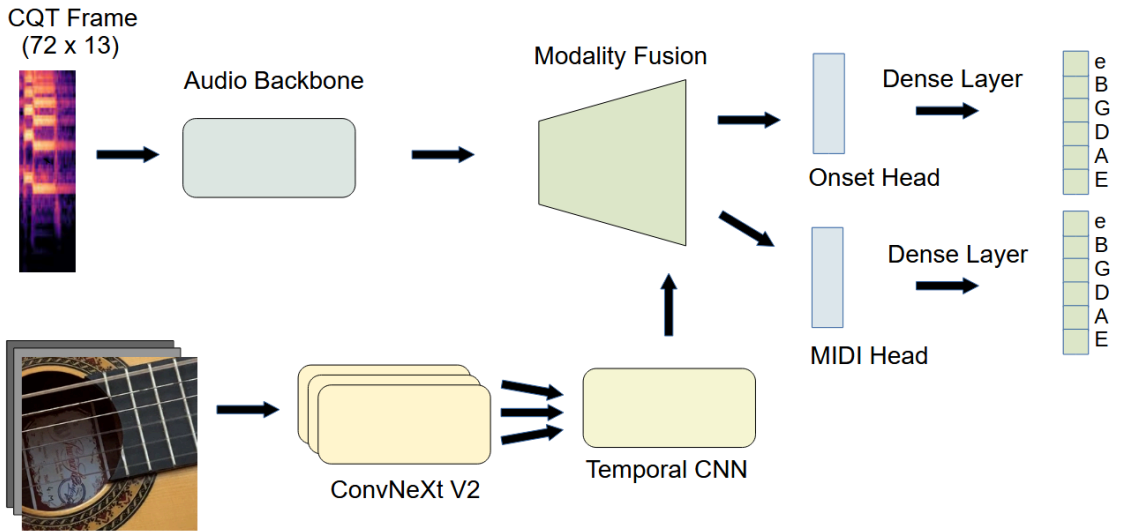


Figure 3.2: Multimodal architecture with cross-attention modality fusion.

4 Dataset

4.1 Public Datasets

For audio-only pretraining, we utilize the GuitarSet dataset [31], which provides hexaphonic audio recordings, precise onset and offset annotations, and pitch labels for solo guitar

performances. GuitarSet contains 360 excerpts (approximately 3 hours total) across multiple playing styles, making it the largest publicly available annotated dataset for guitar transcription. However, GuitarSet does not include synchronized video recordings, limiting its utility for multimodal training.

To the best of our knowledge, no publicly available dataset currently provides synchronized audio–video recordings with string-level onset annotations for guitar transcription. This gap represents a significant bottleneck for developing and evaluating multimodal AMT systems.

4.2 Custom Audio-Video Dataset

To enable multimodal training, we curate a small custom dataset of solo guitar performances with synchronized audio and video. The dataset is recorded under controlled conditions to ensure consistent lighting, camera positioning, and audio quality. While the dataset is not suitable for robust AMT system training, it provides valuable information for testing the theoretical hypothesis.

Recording Setup. Audio is recorded via a built-in smartphone microphone at 44.1 kHz sample rate. Video is captured at 25 fps with 1080p resolution using a smartphone camera positioned to capture the soundboard region.

Annotation Protocol. Annotations are created through a manual pipeline:

- 1 **Initial transcription:** Each performance is manually transcribed into Guitar Pro 5 (.gp5) format, specifying strings and frets.
- 2 **Temporal alignment:** The .gp5 transcription is manually aligned to the actual audio recording using Audacity audio editor [3].
- 3 **Frame-level conversion:** Aligned note events are converted to frame-level labels (23.2 ms resolution) for onset (binary) and pitch (normalized MIDI) targets.

Crucially, our annotation protocol does not require precise offset labels. This design choice significantly reduces annotation time while preserving the information most critical for tablature generation: string, fret, and onset timing.

Dataset Statistics. The current dataset comprises 16 performances (12 recordings for training purposes, 4 for validation and testing), with 20 seconds per recording, on average.² Notably, to provide initial real-world conditions for evaluating the tuning-agnostic nature of our architecture, half of these recordings (8 performances) were played in Drop D tuning, while the

remaining half utilized standard tuning. While modest in scale and diversity compared to GuitarSet, this dataset represents, to our knowledge, the first publicly documented effort to provide synchronized audio–video annotations for guitar tablature transcription with a focus on string motion, rather than the fretboard.

5 Experimental Results

This section presents evaluation results for the audio-only baseline model as well as the multimodal architecture.

5.1 Evaluation Protocol

Metrics. We report three categories of metrics:

- **Onset detection:** Precision, recall, and F1-score computed at the frame level (23.2 ms resolution) per string, averaged across strings.
- **Pitch estimation:** Mean absolute error (MAE) and root mean squared error (RMSE) in semitones, computed only for frames with active onsets.
- **Combined score:** A composite metric that penalizes pitch errors relative to onset quality:

$$\text{Combined} = F1_{\text{onset}} \times \left(1 - \min \left(\frac{\text{MAE}_{\text{pitch}}}{12}, 1 \right) \right) \quad (3)$$

This formulation reflects the practical priority of correct fingering identification over sub-semitone pitch accuracy.

Peak Picking Post-Processing. Frame-level onset probabilities often contain multiple adjacent high-confidence predictions for a single note event. To obtain discrete note onsets, we apply a peak picking algorithm with two parameters:

- **Threshold τ :** The minimum probability for a frame to be considered a candidate onset.
- **Window size w :** The number of neighboring frames to compare when identifying local maxima.

A frame is retained as an onset if its probability exceeds τ and is strictly greater than all neighbors within $\pm \lfloor w/2 \rfloor$ frames. We evaluate multiple threshold values ($\tau \in \{0.3, 0.4, \dots, 0.8\}$) with $w = 5$ frames (116 ms) and report the best-performing configuration.

Datasets and Splits. The audio-only model is pretrained on GuitarSet [31] and evaluated on its official validation and test splits. All results are computed on frame-level predictions before peak picking unless otherwise specified.

5.2 Audio-Only Baseline Results

Frame-Level Evaluation. Table 5.1 summarizes frame-level performance on the GuitarSet validation and test sets.

Table 5.1: Frame-level evaluation results (no post-processing).

Split	Onset F1	Pitch MAE	Pitch RMSE	Combined
Validation	0.439	1.63	2.09	0.380
Test	0.425	1.81	2.31	0.361

The model achieves moderate onset detection performance ($F1 \approx 0.43$) with a pitch estimation error of around 1.6–1.8 semitones. Per-string analysis (Appendix A) reveals a higher F1 score on higher strings, likely due to the superior temporal resolution of higher frequencies.

Peak Picking Evaluation. Applying peak picking post-processing substantially improves onset detection by collapsing adjacent high-confidence frames into discrete events and clipping MIDI values to the nearest integer. Table 5.2 shows results after optimizing the detection threshold τ on the validation set.

Table 5.2: Peak picking evaluation results (threshold $\tau = 0.3$, window $w = 5$).

Split	Onset F1	Pitch MAE	Pitch RMSE	Combined
Validation	0.695	1.62	2.11	0.601
Test	0.660	1.81	2.32	0.560

Key observations:

- 1 **Threshold sensitivity:** The optimal threshold ($\tau = 0.3$) is lower than the default 0.5, indicating that the model produces well-calibrated probabilities where even moderate-confidence predictions often correspond to true onsets.
- 2 **Recall–precision trade-off:** Peak picking increases recall substantially (+37% on validation) at a modest cost to precision (-14%), yielding a net gain in F1-score (Appendix A.2).
- 3 **Pitch stability:** Pitch estimation error remains stable after peak picking, confirming that the regression head learns robust pitch representations independent of onset thresholding.

Comparison to Prior Work. Direct comparison to existing guitar transcription systems is challenging due to differences in evaluation protocols, datasets, and output representations. Moreover, significant improvement from peak picking (+0.26 F1) underscores the importance of task-appropriate post-processing for practical deployment.

5.3 Multimodal Approach Evaluation

We compare four model configurations:

- 1 **Audio-only baseline:** Enhanced Baseline CNN pretrained on GuitarSet, evaluated on the custom dataset (zero-shot transfer).
- 2 **Multimodal + dummy video:** Full multimodal architecture with randomly initialized video frames (sanity check).
- 3 **Multimodal + real video (frozen audio):** Video backbone frozen, audio branch frozen.
- 4 **Multimodal + real video (unfrozen audio):** Video backbone frozen, audio branch unfrozen.

Results. Table 5.3 summarizes the evaluation results on the test set.

Table 5.3: Multimodal evaluation on custom dataset (frame-level, no post-processing).

Configuration	Onset F1	Pitch MAE	Pitch RMSE	Combined
Audio-only baseline	0.309	2.39	2.86	0.248
Multimodal + dummy video	0.000	24.07	24.42	0.000
Multimodal + real video (frozen audio)	0.414	2.27	2.82	0.336
Multimodal + real video (unfrozen audio)	0.336	2.55	3.15	0.265

Key Observations:

- 1 **Visual information provides measurable improvement.** The configuration with frozen audio branch and real video achieves a **+34% relative gain** in onset F1 ($0.309 \rightarrow 0.414$) and **+36% gain** in combined score ($0.248 \rightarrow 0.336$) compared to the audio-only baseline. This result supports our central hypothesis: visual cues from the picking hand help resolve string-fret ambiguity that audio alone cannot disambiguate.
- 2 **Dummy video causes catastrophic failure.** When provided with randomly initialized video frames, the multimodal model collapses to near-zero performance. This confirms that the fusion module does not simply ignore the video input; rather, it actively integrates visual features, and meaningless input disrupts the audio representation. This serves as an important sanity check for the fusion mechanism.

3 Unfreezing the audio branch harms performance on small data. Fine-tuning the audio branch jointly with the video branch leads to degraded results (F1: 0.414 $\rightarrow$ 0.336). We attribute this to overfitting: with only 16 recordings, the model adapts too closely to dataset-specific artifacts (lighting, camera angle, playing style) rather than learning generalizable cross-modal patterns. This underscores the importance of creating a robust multimodal dataset.

Limitations and Interpretation. While these results provide encouraging evidence for the utility of visual information, several caveats must be acknowledged:

- **Dataset scale:** With only few samples, statistical significance is limited. The observed improvements should be interpreted as preliminary evidence rather than definitive proof.
- **Domain shift:** The audio-only baseline was pretrained on GuitarSet, which differs in recording conditions and playing style from our custom dataset. Some performance gap may reflect domain mismatch rather than architectural limitations.
- **Video quality:** All videos were recorded in only two sessions. Real-world deployment would require robustness to variable conditions, which remains untested.
- **Annotation noise:** Manual annotation via Guitar Pro and manual label alignment introduces potential timing errors, particularly for rapid passages.

In summary, these experiments provide the first empirical evidence that visual information from the picking hand can improve guitar tablature transcription, even when training data are scarce. While further validation on larger datasets is needed, the results motivate continued development of multimodal approaches for stringed instrument AMT.

6 Conclusion

This work presents a novel multimodal approach to automatic guitar tablature transcription that addresses the fundamental limitation of audio-only methods: string-fret ambiguity. By integrating visual information from performance videos with audio spectrograms, we demonstrate that complementary modalities have a positive effect on transcription accuracy, even when training data is scarce. Furthermore, our architecture employs a decoupled prediction strategy that separates pitch estimation from physical fingering through continuous MIDI regression rather than

discrete fret classification. This design renders the model inherently adaptable to various tuning configurations without requiring retraining: given any standard or alternative tuning as side information, string-fret assignments can be derived post-hoc through straightforward arithmetic.

Based on our findings, we identify three concrete directions for future research:

- 1 **Scale the dataset:** Expanding the custom dataset to 50–100 recordings would enable more reliable evaluation, reduce overfitting risk during fine-tuning, and support unfreezing the audio branch for end-to-end optimization.
- 2 **Optimize the audio branch:** Investigating various audio input dimensions and scaling the model’s trainable parameters may further enhance performance toward state-of-the-art results.
- 3 **Develop an advanced note-level framework:** Utilizing the proposed architecture as an encoder enables the development of a sophisticated transcription system capable of directly mapping audio-visual signals to symbolic representations.

It is crucial to acknowledge the empirical limitations of this study. Without access to specialized hardware such as hexaphonic pickups to automate label generation, curating synchronized audio-video data proved exceptionally labor-intensive, requiring approximately 40 minutes of expert time for each minute of audio. Consequently, the dataset remains statistically underpowered (16 recordings), meaning that while the observed improvements in multimodal fusion are promising, they should be viewed as a proof-of-concept rather than a definitive, large-scale validation. The overfitting observed when unfreezing the audio branch directly reflects this data scarcity. Nonetheless, the pitch-regression formulation demonstrates practical viability for alternative tunings without model retraining.

Beyond guitar transcription, this work contributes to the broader field of multimodal music understanding. The proposed architecture demonstrates that visual information can complement audio signals, even when the primary task (pitch estimation) is traditionally considered auditory. This opens possibilities for other stringed instruments (bass, violin, ukulele) where positional ambiguity similarly limits audio-only approaches.

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A Detailed Per-String Metrics

This appendix provides comprehensive per-string evaluation metrics for all experimental configurations discussed in Section 5. Metrics are reported at the frame level (23.2 ms resolution) unless otherwise specified.

A.1 Audio-Only Baseline: GuitarSet Validation Set

Table A.1 shows per-string metrics for the enhanced baseline CNN evaluated on the GuitarSet validation split, both before and after peak picking post-processing.

Table A.1: Per-string metrics: Audio-only baseline on GuitarSet validation set.

String	Frame-Level			Peak Picking ($\tau = 0.3$)		
	Prec.	Rec.	F1	Prec.	Rec.	F1
E4 (s0)	0.584	0.445	0.505	0.481	0.687	0.565
B3 (s1)	0.486	0.457	0.471	0.663	0.785	0.719
G3 (s2)	0.515	0.367	0.428	0.692	0.783	0.734
D3 (s3)	0.448	0.392	0.418	0.642	0.813	0.718
A2 (s4)	0.436	0.432	0.434	0.609	0.783	0.685
E2 (s5)	0.419	0.346	0.379	0.506	0.780	0.614
Average	0.481	0.406	0.439	0.625	0.783	0.695

A.2 Audio-Only Baseline: GuitarSet Test Set

Table A.2 presents results on the held-out GuitarSet test split.

Table A.2: Per-string metrics: Audio-only baseline on GuitarSet test set.

String	Frame-Level			Peak Picking ($\tau = 0.3$)		
	Prec.	Rec.	F1	Prec.	Rec.	F1
E4 (s0)	0.510	0.368	0.428	0.489	0.640	0.554
B3 (s1)	0.439	0.450	0.445	0.593	0.752	0.663
G3 (s2)	0.507	0.377	0.432	0.646	0.801	0.715
D3 (s3)	0.406	0.333	0.366	0.636	0.811	0.713
A2 (s4)	0.403	0.446	0.423	0.524	0.731	0.610
E2 (s5)	0.541	0.397	0.458	0.504	0.700	0.586
Average	0.468	0.395	0.425	0.584	0.758	0.660

A.3 Audio-Only Baseline: Custom Dataset (Zero-Shot Transfer)

Table A.3 shows results when the GuitarSet-pretrained model is evaluated on the custom audio-video dataset without fine-tuning. This represents a zero-shot transfer scenario with domain shift.

Table A.3: Per-string metrics: Audio-only baseline on custom dataset (zero-shot).

String	Precision	Recall	F1	Pitch MAE
E4 (s0)	1.000	0.276	0.432	2.51
B3 (s1)	0.238	0.172	0.200	2.16
G3 (s2)	0.368	0.259	0.304	1.79
D3 (s3)	0.333	0.208	0.256	1.61
A2 (s4)	0.235	0.364	0.286	2.97
E2 (s5)	0.462	0.316	0.375	3.23
Average	0.439	0.266	0.309	2.39

A.4 Multimodal Experiments: Custom Dataset

Table A.4 provides per-string metrics for the four multimodal configurations evaluated on the custom dataset.

Table A.4: Per-string metrics: Multimodal configurations on custom dataset.

Config / String	Prec.	Rec.	F1	Pitch MAE
<i>Audio-only baseline (zero-shot)</i>				
E4	1.000	0.276	0.432	2.51
B3	0.238	0.172	0.200	2.16
G3	0.368	0.259	0.304	1.79
D3	0.333	0.208	0.256	1.61
A2	0.235	0.364	0.286	2.97
E2	0.462	0.316	0.375	3.23
<i>Multimodal + dummy video (sanity check)</i>				
All	0.000	0.000	0.000	24.07
<i>Multimodal + real video (frozen audio) [Best]</i>				
E4	0.609	0.483	0.539	3.05
B3	0.441	0.517	0.476	1.83
G3	0.417	0.185	0.256	1.88
D3	0.500	0.458	0.478	2.14
A2	0.184	0.636	0.286	2.73
E2	0.583	0.368	0.452	1.97
<i>Multimodal + real video (unfrozen audio)</i>				
E4	0.714	0.517	0.600	2.45
B3	0.390	0.552	0.457	2.12
G3	0.286	0.148	0.195	2.03
D3	0.333	0.333	0.333	2.73
A2	0.111	0.091	0.100	3.26
E2	0.800	0.211	0.333	2.73